

Validation of the LUMEN EcosimPro Model with Hot-Fire Test Data

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Abstract

LUMEN (Liquid Upper stage deMonstrator Engine) is a LOX/LNG expander-bleed type engine with two turbopumps in the 25 kN thrust range. The project was started in 2017 at the Institute of Space Propulsion of the German Aerospace Center (DLR). Since then, several test campaigns have been carried out for the thrust chamber assembly and each turbopump while the first hot-fire test of the whole engine was successfully conducted in 2024. The experimental data collected during each test campaign provides an ideal basis for validating transient simulation models at component and system level. The LUMEN simulation model is built in EcosimPro with the European Space Propulsion System Simulations (ESPSS) toolkit. In this work, the models for the combustion chamber, injector, cooling channels, turbines and pumps are first validated individually. Subsequently, the entire LUMEN model is validated using hot-fire test data. The results show that critical parameters, such as the combustion chamber pressure, the fuel injection temperature and the rotational speed of each turbopump, are modelled with high accuracy. Overall, the mean average percentage error is below 5 % for each quantity. The results of this work allow fast, simple and risk-free predictions of the system behaviour of LUMEN. This is particularly important in the preparation of future test campaigns and for developing intelligent controllers and health monitoring systems.

1. Introduction

The Institute of Space Propulsion at the German Aerospace Center (DLR) initiated the LUMEN (Liquid Upper stage deMonstrator Engine) project in 2017, aiming to develop system-level expertise for the operation of liquid rocket engines. LUMEN, as shown in Figure 1, is an expander-bleed rocket engine which operates with liquid oxygen (LOX) and liquefied methane (LNG). Each propellant is pressurised by its own turbopump. LNG is partially used for the regenerative cooling of the main combustion chamber (MCC) in a counterflow-arrangement. The heated LNG mass flow is then split between a mixer, where it is used to elevate the temperature of the fuel before injection into the MCC, and the cooling channels of the nozzle extension (NE). The NE coolant is then used to drive the turbines before being discarded. LUMEN generates 25 kN thrust in its nominal operating point at a combustion chamber pressure p_{cc} of 60 bar and a mixture ratio ROF of 3.4. In its latest configuration, LUMEN uses three electrically actuated control valves for setting the operating point: the turbine fuel valve (TFV), the turbine oxidiser valve (TOV) and the fuel control valve (FCV).

Within the LUMEN project the thrust chamber assembly (TCA) and each turbopump were first developed and tested individually before hot-fire test campaigns of the whole engine were successfully conducted at the European research and technology test bench P8.3 in Lampoldshausen in 2024 and 2025 [8, 18]. Two combustors with similar pressure loss and heat transfer characteristics were developed and tested in the project. The first combustor is conventionally manufactured and uses a CuCrZr inner liner with axially milled and galvanically closed cooling channels. Laser powder bed fusion (LPBF) was used to manufacture the second single piece combustor liner. Details of the development and manufacturing process of the conventional combustor and the additive manufactured combustor are provided in Haemisch et al. [11] and Suslov et al. [16, 17]. The injector head features 42 shear-coaxial elements. Characterisation of the combustion process and verification of flame anchoring are achieved by means of a measurement ring equipped with temperature and pressure sensors as well as optical probes.

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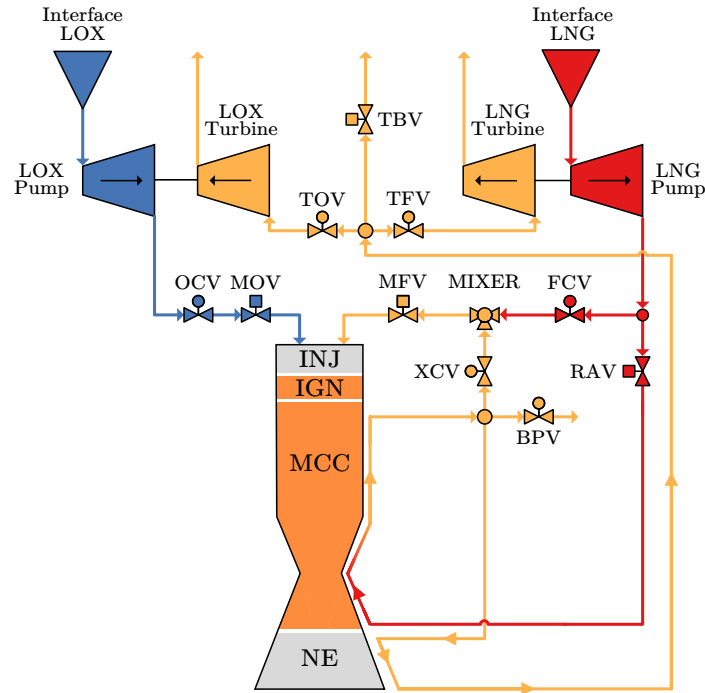


Figure 1: Flow scheme of LUMEN.

The fuel and oxidiser turbopump (FTP and OTP), as shown in Figure 2, use a single-stage centrifugal pump and supersonic impulse turbine. The design has been carried out using in-house tools [18]. Details of the pump and turbine design are given in Hahn et al. [4, 5]. To increase the life expectancy of the bearing system, oil lubrication with commercial of the shelf bearings is used. With a minimum expected life of 1000 h, the bearing system exceeds the life time of typical turbopump designs significantly. The maximum rotational speed of the OTP and FTP is 28 000 rpm and 50 000 rpm, respectively.

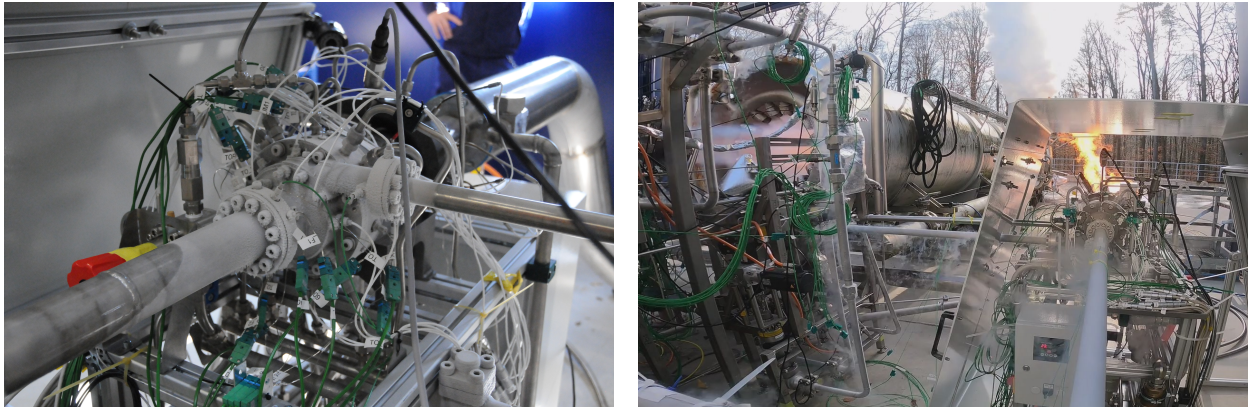


Figure 2: OTP after sub-component test (left) and FTP during hot-fire test (right).

In the current shift of the space industry towards re-usability and cost reduction, advanced technologies such as intelligent control and health monitoring play a vital role. Given its modular design, extensive instrumentation and wide throttling range, LUMEN is ideally suited for testing and developing control and health monitoring systems under realistic conditions. The development of those technologies requires a transient simulation model and a detailed understanding of the engine. The sub-component and hot-fire tests, which were carried out within the LUMEN project provide a large database for building and validating the required simulation model. This paper presents the detailed modeling of the LUMEN simulation model using the European Space Propulsion System Simulation (ESPSS) toolkit of EcosimPro and its validation using experimental data. For this, an overview of the test activities which were carried out within the LUMEN project is given first. After, the sub-component modelling and validation are described. Finally, the complete simulation model is validated using experimental data from the latest hot-fire test campaign.

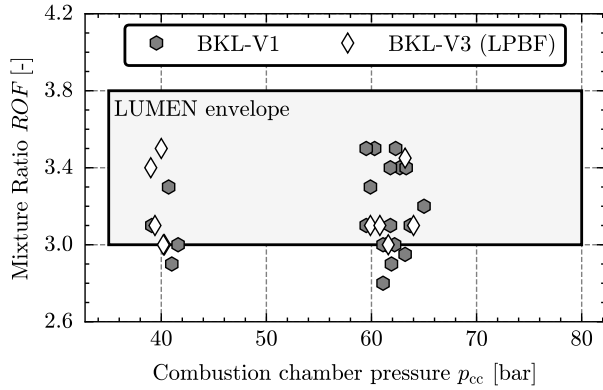


Figure 3: Steady-state load points achieved during TCA test campaigns for both combustors [11, 16, 17].

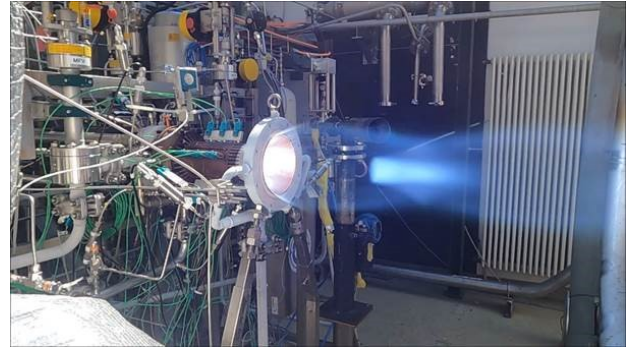


Figure 4: Thrust chamber during hot fire testing.

2. Test activity overview

Design qualification and characterisation of each subsystem were achieved over multiple test campaigns. This includes two test campaigns for the TCA, two test campaigns for each turbopump and two hot-fire test campaigns of the whole engine. In the following section an overview of these test campaigns is provided.

2.1 Thrust chamber assembly

The TCA test campaigns were conducted at the P8 test facility in Lampoldshausen, focusing on the characterisation of the injector head, combustion chamber, nozzle extension, regenerative cooling system and control valves [11, 16]. In addition to qualifying the design of both combustors, several goals were successfully achieved:

- Characterisation of the injector head pressure loss
- Determination of the combustion efficiency and minimum required cooling channel mass flow
- Performance characterisation with fuel injection temperatures between 210 K and 290 K

In total, 27 tests have been conducted with an accumulated test duration of 24 minutes. As shown in Figure 3, representative steady-state load points have been achieved for both combustors within the LUMEN operational envelope. More details on the experimental setup and results can be found in Börner et al. [3], Hardi et al. [13], Haemisch et al. [11] and Suslov et al. [16, 17].

2.2 Turbopumps

Multiple test campaigns have been conducted for the characterisation of the performance of both turbopumps, the qualification of the mechanical and thermal design and the development of a reliable chill-down procedure [18]. For each turbopump unit individual test campaigns have been conducted. Following the initial water test of the OTP, cryogenic test campaigns were conducted at the P8.3 test facility in Lampoldshausen. In those test campaigns, the turbine was driven with gaseous nitrogen and the pump discharge pressure and mass flow were set by a downstream control valve and orifice.

During the OTP test campaign at the P8.3 test facility, the employment of a neural network based controller for tracking reference trajectories of the pump discharge pressure and pump mass flow rate was successfully demonstrated for the first time by Dresia et al. [7]. The intelligent controller achieved high tracking accuracy over an accumulated runtime of 500 s with errors below 1 %. In addition, fault diagnostic methods were successfully tested in the FTP test campaign by Kurudzija et al. [14]. Within this work, semi-empirical models were derived which accurately describe the relationship between the pump discharge pressure, mass flow rate, rotational speed and the actuator position of the FCV. The mathematical models were successfully employed at the test bench P8.3 and achieved errors below 3 %. In post-processing, a fault in the rotational speed sensor with a detection delay below 1 s could be detected. This demonstrated the capability of using those models for fault diagnostic in future health monitoring systems. Over five test campaigns, the OTP and FTP accumulated a runtime of 88 minutes and 44 minutes, respectively.

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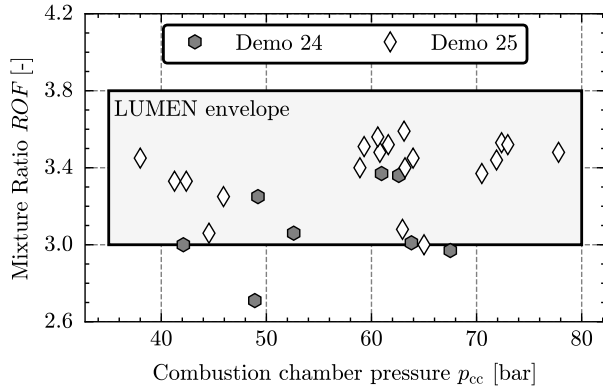


Figure 5: Steady-state load points achieved during the hot-fire test campaigns [8].

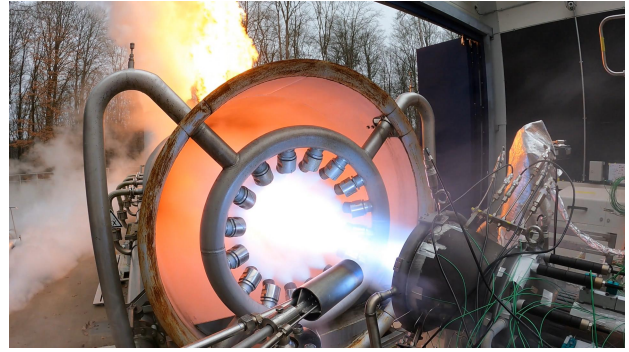


Figure 6: LUMEN hot fire testing at test bench P8.3.

2.3 Engine

Following the sub-component tests, the entire LUMEN demonstrator engine has been successfully tested and put into service at the test bench P8.3 in 2024 as shown in Figure 6. Within this test campaign, the ignition sequence was initially established, with the subsequent characterisation of the operational envelope. With the objective of further expanding the operational envelope, testing the throttling capabilities and optimising sub-components of LUMEN, a second test campaign was conducted in 2025. In comparison to the preceding test campaign, the LOX impeller has been improved, an injector head which features swirl elements on the inner row and shear-coaxial elements on the outer row has been utilised, the chill-down process has been improved and the system pressure losses have been reduced [8]. Over the course of both test campaigns, numerous significant milestones have been achieved:

- Characterisation of the engine for combustion chamber pressures between 38 bar and 78 bar.
- Demonstration of the repeatability of the engine conditioning and ignition sequences.
- Demonstration of high durability of the TCA over 27 ignitions and test operation exceeding 24 minutes.
- Validation of the thermal design of the turbopumps during long-duration tests.
- Establishment of a large database on LOX/LNG injection over a wide fuel injection temperature and combustion chamber pressure range.
- Demonstration of the self-start capability without relying on a GN_2 turbine starter.
- Demonstration of a neural network controller for thrust and mixture ratio control.
- Demonstration of operation along a sequence analogous to a moon landing.

In total, 17 ignitions of LUMEN have been successfully performed in both test campaigns with an accumulated hot-fire test duration of over 20 minutes [8]. As shown in Figure 5, the whole operational envelope of LUMEN has been successfully characterised in both test campaigns. More details are provided in Dresia et al. [8].

3. Fundamentals

Following the establishment of a large database for each sub-component and the entire engine, accurate transient simulation models can be developed and validated across the whole operational envelope of LUMEN. The simulation models are built in EcosimPro 7.0.10. In the following section an overview of the measurement uncertainties is provided, followed by a description of the mathematical fundamentals of EcosimPro.

Table 1: Measurement accuracy and uncertainties.

Measurement	Measurement range	Max. uncertainty
Temperature	106 – 313 K	± 2.5 K
	233 – 648 K	± 1.5 K
Mass flow rate	0 – 10 kg/s	± 0.03 kg/s
Volumetric flow rate	0 – 12 l/s	± 3 %
Pressure	0 – 250 bar	± 1 %
Rotational speed	0 – 200 000 rpm	± 0.2 %

3.1 Measurement uncertainties

For characterising the operational condition of the components, extensive instrumentation is used within each test campaign of the LUMEN project. This includes measurements of pressures, temperatures, mass flow rates, rotational speeds and vibrations. As the fluid properties are varying within the engine, different types of sensors and measurement ranges are used depending on the expected conditions. Thermocouples used for measuring cryogenic and non-cryogenic temperatures are mostly of type K and introduce an uncertainty of ± 2.5 K between 106 K and 313 K and ± 1.5 K between 233 K and 648 K, respectively. A broad range of different static pressure sensors is used within LUMEN with the maximum uncertainty of the static pressure sensors being ± 1 % from the full span measurement range. For measuring the injection mass flow rate on the LOX and LNG side, Coriolis flow meter are used. The maximum uncertainty is ± 0.03 kg/s. Turbine flow meter are used for the other flow rate measurements. Using temperature and pressure measurements to estimate the fluid density with lookup tables, the volumetric flow rate is converted to a mass flow rate measurement in the MCC. The uncertainty of the whole measurement chain is ± 3 % [1, 12]. The rotational speed of the turbopumps is measured with an inductive sensor which has a published linearity of ± 0.2 %. The considered uncertainties are summarized in Table 1.

3.2 Mathematical modeling

EcosimPro solves systems of differential algebraic (DAE) and ordinary differential equations (ODE) and is able to handle discontinuities introduced by discrete events such as valve switching or phase transition [9]. It uses an object-oriented modelling approach in which complex physical systems, e.g. the LUMEN engine, are built with more elementary components, e.g. pipes or valves. Each component is defined by its own system of equations and ports through which mass, momentum and energy are exchanged. By defining boundary conditions, EcosimPro automatically assembles a global system of equations. Denoting $\mathbf{x}(t) \in \mathbb{R}^n$ as the vector of explicit algebraic variables and $\mathbf{y}(t) \in \mathbb{R}^m$ as the vector of semi-state and implicit algebraic variables, the mathematical problem can be stated as

$$\dot{\mathbf{x}} = \mathbf{G}(t, \mathbf{y}, \mathbf{x}, \mathbf{p}) \quad \text{and} \quad \mathbf{F}(t, \mathbf{y}, \dot{\mathbf{y}}, \mathbf{x}, \mathbf{p}) = 0, \quad (1)$$

where $\mathbf{p}$ collects the non-explicit variables. For solving the system of equations, several transient solvers with variable-step integration can be used. In this work, the sparse CVODE solver with backward differentiation formula is used.

4. Component modeling and validation

The LUMEN simulation model is built using the European Space Propulsion System Simulation (ESPSS) toolkit version 3.8.1 of EcosimPro. ESPSS provides a comprehensive collection of components for the simulation of rocket engines and satellites [10]. It includes libraries for calculating the thermodynamic and transport properties for perfect and real fluids as well as mixtures of fluids and chemicals. Propulsion-specific components such as turbines, pumps, combustion chambers, pipes and valves are implemented in ESPSS libraries. Governing equations for mass continuity, momentum and energy conservation are enforced within those components. Each component is characterised by a unique set of parameters and equations. The details of the most relevant components used in the LUMEN model are described and validated in the following. As LUMEN's components have been updated progressively, only stationary load points from the latest hot-fire test campaign (see Figure 5) are used for validation. The initial modeling of each component and the complete LUMEN model has been conducted by Dresia [6].

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4.1 Injector

For modelling the injector, the standard ESPSS component is used [10]. The mass flow $\dot{m}$ is calculated dynamically by balancing the momentum using

$$0.5 I \frac{d\dot{m}}{dt} = (p + 0.5\rho v^2)_{\text{in}} - (p + 0.5\rho v^2)_{\text{out}} - 0.5 \frac{\zeta G^2}{\rho_{\text{in}}}. \quad (2)$$

Here I is the inertia of the injector, p the pressure, ρ is the density, v the velocity of the fluid, ζ the effective pressure loss coefficient and G the mass flow per unit area. The validation of the modeling of the injector is conducted by comparing the pressure drop across each injector. The pressure drop is a function of the mass flow through the injector, the fluid density and the area discharge coefficient $(C_{dA})_{\text{inj}}$, defined as

$$\Delta p = \frac{\dot{m}_{\text{inj}}^2}{2\rho (C_{dA})_{\text{inj}}^2}. \quad (3)$$

To validate the mathematical models for each injector, the mass flow, inlet temperature and pressure are imposed as boundary conditions and the outlet pressure is compared against experimental data. As shown in Figure 7, the model aligns well with experimental data for both injectors. At the given temperatures, LOX can be assumed to be incompressible, the pressure drop changes therefore approximately quadratically with an increasing mass flow. The compressibility effects on the LNG side are also accurately modelled. In total, the pressure loss across the LOX and LNG injector is modeled with errors below 4 % and 5 %, respectively.

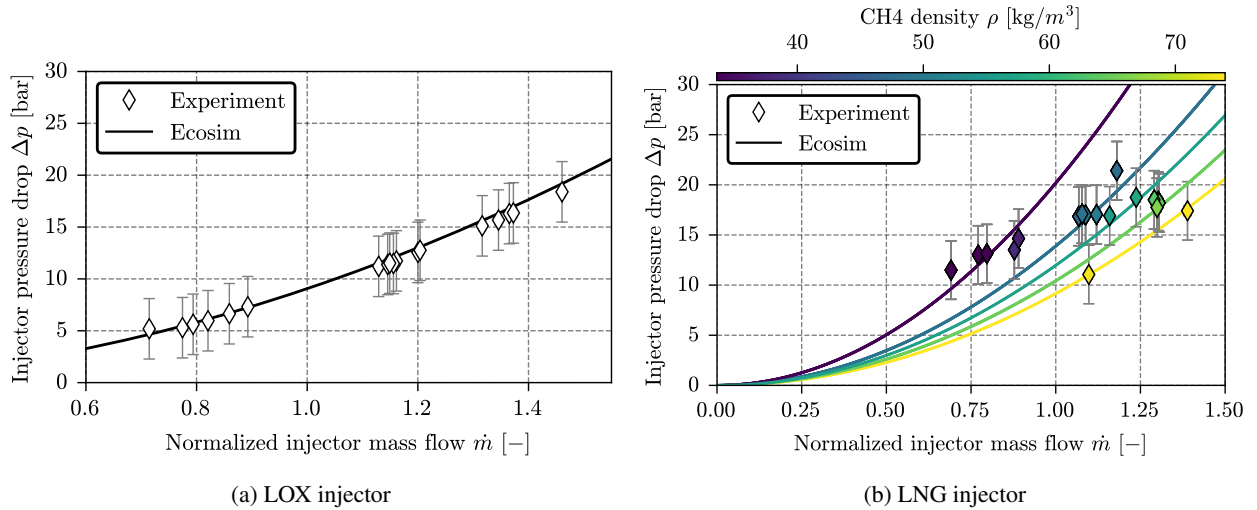


Figure 7: Comparison of the predictions of the injector components with experimental data.

4.2 Cooling channels

The regenerative cooling of LUMEN is modeled using two separate cooling channels for the combustion chamber and nozzle extension, respectively. ESPSS models the cooling channels using a central fluid channel with surrounded wall components. Fluid dynamic effects, such as pressure loss due to internal friction and variations in the velocity of the cooling medium, as well as thermodynamic effects resulting from heat transfer with the cooling channel walls, are modeled within the central fluid channel. This represents a resistive flow element, thermally interacting with the metal structures through a one-dimensional thermal port. The heat transfer inside the metal structure of the cooling channel due to conductivity is modeled using the wall components. The detailed description of the mathematical model of the cooling channels is given in [10].

To validate the modeling, the cooling channels are simulated alongside the combustion chamber. The heat transfer is modeled by a modified Bartz correlation, taking into account the stronger influence of the fuel injection temperature and combustion chamber pressure for LUMEN [6]. The mass flow, inlet temperature and pressure for both components are imposed as boundary conditions. Similarly to the injector component, the cooling channel is validated by comparing the pressure drop as well as the integral heat flux to experimental steady-state points. As

shown in Figure 8 the model aligns well with experimental data for both the pressure drop and integrated heat flux. The maximum error in the pressure drop of the combustion chamber and nozzle extension cooling channels is 1.4 bar and 1.7 bar, respectively. For the integrated heat flux, the maximum error in both cooling channels is below 6 %.

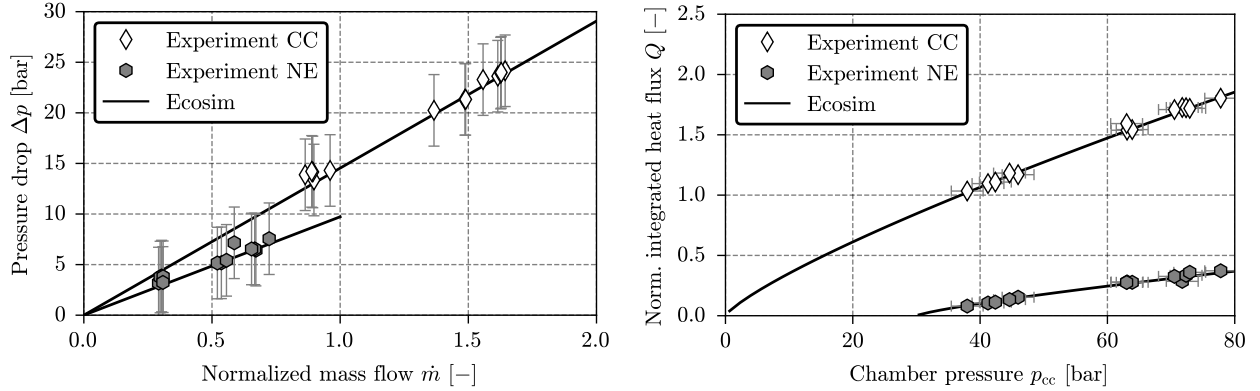


Figure 8: Comparison of the model predictions for the pressure drop and integral heat flux with experimental data.

4.3 Pumps

For modelling the behavior of the pump, ESPSS provides different components [10]. In the early design stage, a generic pump component can be used in which the performance curves are calculated based on the design point and specific speed of the pump using Suter curves. To enable accurate modelling of the off-design behavior, performance curves for head rise and torque can be provided in the standard pump component. In this work, the standard component is used and the initial performance curves, provided by Hahn et al. [15], are modified to match experimental data.

The rotational speed ω of the pump is calculated dynamically by balancing the torque T at the shaft with the torque consumed of the pump under consideration of the inertia of the pump I_p

$$I_p \frac{d\omega}{dt} = T_{\text{shaft}} - T. \quad (4)$$

For energy conservation, the product of enthalpy h and mass flow is balanced under consideration of the added mechanical work to the fluid at the inlet

$$\dot{m}_p h_{\text{out}} = \dot{m}_p h_{\text{in}} + T \omega. \quad (5)$$

Neglecting cavitation, the mass flow is calculated dynamically with

$$I_{\text{fluid}} \frac{d\dot{m}_p}{dt} = (p + 0.5\rho v^2)_{\text{in}} - (p + 0.5\rho v^2)_{\text{out}} + g \rho_{\text{in}} H_T, \quad (6)$$

where I_{fluid} is the inertia of the fluid and H_T is the total dynamic head. Both characteristic curves are based on the outlet flow coefficient ϕ^+ which is defined as

$$\phi^+ = \frac{\dot{m}_p}{\rho_{\text{in}} \omega}. \quad (7)$$

The head coefficient ψ^+ describes the pump head rise normalized by the rotational speed and is modeled as a function of the outlet flow coefficient

$$\psi^+ = \frac{p_{\text{out}} - p_{\text{in}}}{\rho_{\text{in}} \omega^2} = f(\phi^+). \quad (8)$$

In Belyaev [2] the relationship between the flow and head coefficient is expressed as an polynomial relationship

$$\psi^+ = a + b \phi^+ - c \phi^{+2}, \quad (9)$$

where the polynomial coefficient can be derived from the pump design point. In this work, the coefficients are estimated with experimental data for both pumps. As demonstrated in Figure 9 (a), the polynomial relationship aligns well with experimental data of the FTP. However, the experimental data of the OTP indicates that the rotational speed has a stronger influence on the head rise. To account for this behavior, a linear correction term based on the rotational speed is introduced for the head coefficient

$$\psi_{\text{corr}}^+ = \psi^+ + f(\omega) = f(\phi^+, \omega). \quad (10)$$

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With this correction, the experimental data of the OTP can be accurately reproduced over a wide flow coefficient range, as shown in Figure 9 (b). For further validation, an isolated simulation environment is built for each pump. Each component is simulated individually with imposed boundary conditions from experimental data. In this setup the inlet temperature and pressure, rotational speed and mass flow are imposed as boundary conditions while the outlet pressure is used for validation. The maximum errors of the FTP and OTP lies below 1 % and 3 %, respectively. This demonstrates that the provided relationship for the head and flow coefficient is modeled accurately for both pumps.

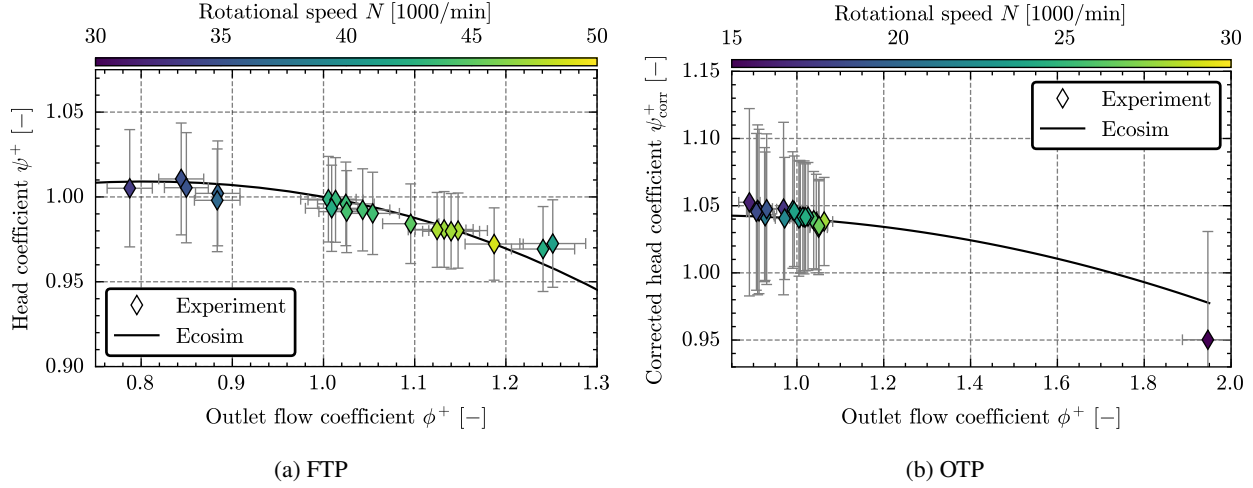


Figure 9: Relationship between the flow coefficient and the head coefficient for the FTP and the corrected head coefficient for the OTP. All coefficients are normalized against the nominal operational point.

The second pump curve describes the torque coefficient λ^+ as a function of the flow coefficient

$$\lambda^+ = \frac{T}{\rho_{in} \omega^2} = f(\phi^+), \quad (11)$$

where T is the consumed torque of the pump. During the LUMEN test campaigns, no torque measurement has been conducted. For estimating the torque coefficient, the fluid enthalpy h at the inlet and outlet is estimated using temperature and pressure measurements. Using the definition of the isentropic efficiency

$$\eta_p = \frac{h_{out,s} - h_{in}}{h_{out} - h_{in}} \quad (12)$$

and rearranging the torque, head and flow coefficient as

$$\eta_p = \frac{\phi^+ \psi^+}{\lambda^+}. \quad (13)$$

the torque coefficient can be estimated using the available measurements. As shown in Figure 10, the torque coefficients for both pumps can be estimated such that the isentropic efficiencies aligns with experimental data. For validating the torque curve, the same simulation setup as for the head curve is used and the outlet temperature is compared against experimental data. The maximum error for the FTP and OTP is 1.6 K and 1.8 K, respectively. This indicates that the torque for both pumps is modeled accurately. However, it should be mentioned that the fluid temperature increase between inlet and outlet of the pumps is in the order of 10 K and subject to both measurement uncertainties of the cryogenic thermocouples as well as heat ingress through the pump structure.

4.4 Turbine

Analogous to the pump, ESPSS provides multiple components for the turbine [10]. The generic turbine calculates the performance map based on the rated conditions. In the standard turbine component, this map is an input parameter which allows an accurate modeling of the off-design behavior. In this work, the standard component is used and the initial performance curves, provided by Hahn et al. [15], are modified to match experimental data.

The mechanical balance and energy conservation of the turbine is similar to the pump (see Equation 4 and 5) with the difference that the mechanical work is added to the fluid at the outlet. As LUMEN uses supersonic impulse

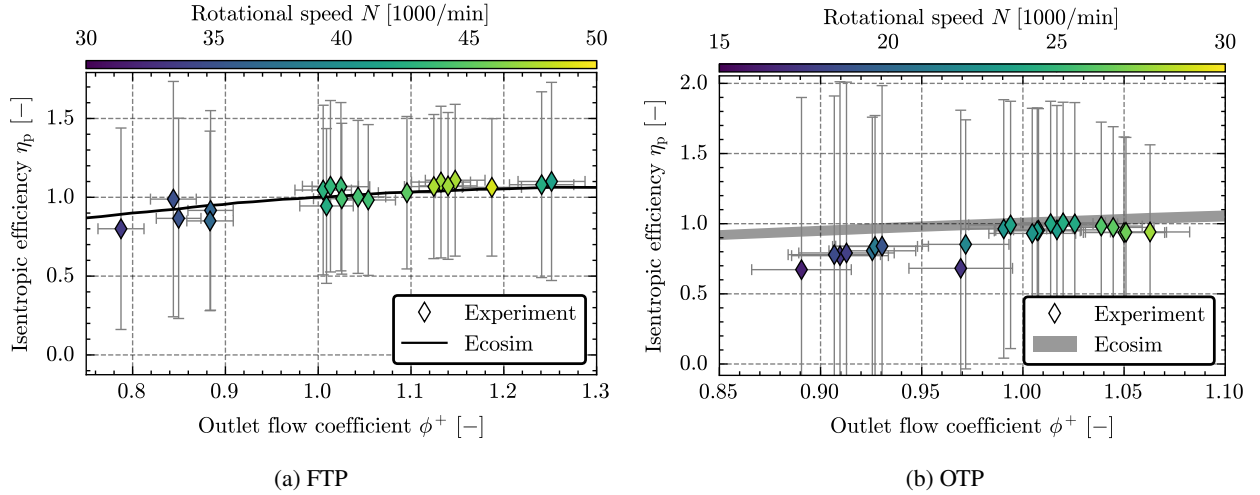


Figure 10: Relationship of the isentropic pump efficiency and flow coefficient for both turbopumps. As the efficiency of the OTP is modeled as a function of the corrected head coefficient, which is dependent on rotational speed, the colored region shows the efficiency for the boundaries of the rotational speed in the operational envelope of LUMEN. All quantities are normalized against the nominal operational point.

turbines, the inlet mass flow is computed assuming choked flow at the throat of the inlet nozzles

$$I_{\text{fluid}} \frac{d\dot{m}_t}{dt} = (p + 0.5\rho v^2)_{t,\text{in}} - (p + 0.5\rho v^2)_{t,\text{out}} - \frac{0.5(\zeta + \zeta_{\text{crit}}) G^2}{\rho_{\text{in}}}. \quad (14)$$

Here ζ is the pressure loss coefficient and ζ_{crit} is the sonic flow correction factor. For the construction of the turbine performance map, coefficients describing the blade speed, pressure ratio and torque are used. The speed coefficient N_{red} relates the blade speed to the velocity of the working fluid as

$$N_{\text{red}} = \frac{r_t \omega}{a_{\text{out}}}, \quad (15)$$

where r_t is the characteristic blade tip radius and a_{out} is the speed of sound at the outlet. The total pressure ratio π_t is defined as the ratio of the total pressure at the inlet and outlet

$$\pi_t = \frac{p_{t,\text{in}}}{p_{t,\text{out}}}. \quad (16)$$

The specific torque τ is a function of the speed coefficient and total pressure ratio and can be expressed as

$$\tau = \frac{T}{r_t \dot{m}_t a_{\text{out}}} = f(N_{\text{red}}, \pi_t). \quad (17)$$

As mechanical losses are considered, the turbine performance maps are estimated by simulating the balancing of turbine generated and pump consumed power. For this, the turbines and pumps are simulated in an isolated simulation environment using boundary conditions for the mass flow, rotational speed, inlet temperature and pressure from experimental steady-state points. Finally, the pump performance curves described in Section 4.3 are fixed and the performance map of the turbine is updated iteratively until a power balance is approximately reached. The results of both turbopumps are shown in Figure 11. The maximum power imbalance for the FTP and OTP is 6 % and 5 %, respectively. This demonstrates that an approximate power balance over the entire range of experimental data could be reached using the estimated turbine performance map.

5. System analysis

As demonstrated in the previous section, the modeling of the individual components is accurate. However, coupling effects and propagation of unavoidable modeling errors can result in an inaccurate complete model. For validation, all components are integrated into a complete model of LUMEN. The predictions of this model are compared against experimental data from a single test run after the ignition and start-up of the engine. This test run has been specifically

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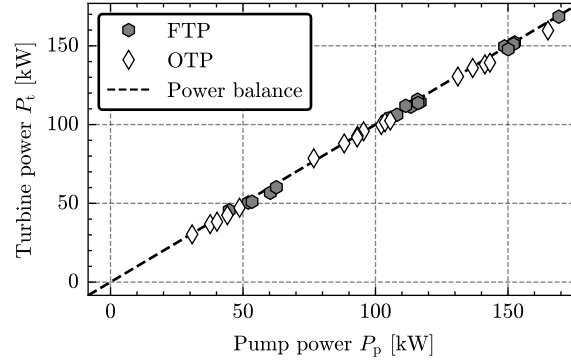


Figure 11: Power balance for both turbopumps.

designed for validation purposes, as it covers the entire operational range of LUMEN, including steps and ramps as transients. The temperature and pressure conditions at the interface of the test bench for LOX and LNG are imposed as inlet boundary conditions. Outlet boundary conditions are set to standard temperature and pressure. The control signals of TFV, TOV and FCV are also imposed as boundary condition. The comparison of the model predictions and the experimental data for various pressures and temperatures, the rotational speed of both turbopumps and the mixture ratio is shown in Figure 12. The mean average percentage error (MAPE) over the entire test duration and the maximum absolute error of steady-state points are reported in Table 2.

Overall, the model is able to accurately reproduce the combustion chamber pressure and mixture ratio with a MAPE of 2.4 % and 3.5 %, respectively. With a maximum error of 2.4 bar or 3.3 %, the steady-state combustion chamber pressure is modeled with high accuracy. Generally, the model captures pressures with high accuracy both in steady-state and during transients. The oxidiser pump outlet pressure is modeled the poorest with a maximum absolute error and MAPE of 7.4 % and 4.1 %, respectively. Despite the considerable variation, this is solely 3.2 bar. This indicates that the model is able to replicate the pressure drop across the cooling channels, valves, pipes and injectors. The same trend is evident for the temperatures. The maximum steady-state error is reached for the cooling channel outlet temperature with 6.0 %, respectively 25.4 K. As the rate of change of the temperatures has a high time scale, the transient dynamics can be effectively evaluated using the MAPE. The maximum MAPE is 2.1 % for the cooling channel outlet temperature which indicates that the model is able to accurately capture the temperature increase in the feedlines and cooling channels. Further, the model is able to replicate the rotational speed of both turbopumps with steady-state errors of approximately 3.0 % and a MAPE of 1.8 %. Given the high modeling accuracy in the temperatures, pressures and rotational speeds, it can be concluded that the torque relationships for the pumps and turbines, which were determined with high uncertainty, reflect the correct trend. The model is also able to replicate the mass flows accurately over the whole test duration. This is indicated by the maximum steady-state error and MAPE of the mixture ratio of 5.2 % and 3.5 %, respectively.

Table 2: Prediction error of the LUMEN simulation model for different pressures, temperatures, the rotational speed of both turbopumps and the mixture ratio.

Parameter	MAPE ¹	Max. percentage error ²	Max. absolute error ²
Combustion chamber pressure p_{cc}	2.4 %	3.3 %	2.4 bar
Fuel pump outlet pressure $p_{f,out}$	2.8 %	4.8 %	5.2 bar
Oxidiser pump outlet pressure $p_{o,out}$	4.1 %	7.4 %	3.2 bar
Fuel injection temperature $T_{inf,f}$	1.9 %	4.8 %	15.1 K
Cooling channel outlet temperature $T_{rc,out}$	2.1 %	6.0 %	25.4 K
Turbine inlet temperature $T_{t,in}$	2.0 %	4.0 %	23.2 K
Rotational speed FTP N_{FTP}	1.5 %	3.1 %	1292 rpm
Rotational speed OTP N_{OTP}	1.8 %	2.6 %	684 rpm
Mixture ratio ROF	3.5 %	5.2 %	0.2

¹ Calculation over the test duration after ignition and start-up.

² Only steady-state points are considered.

Overall, the high accuracy of the simulation model over the entire operational envelope of LUMEN makes it

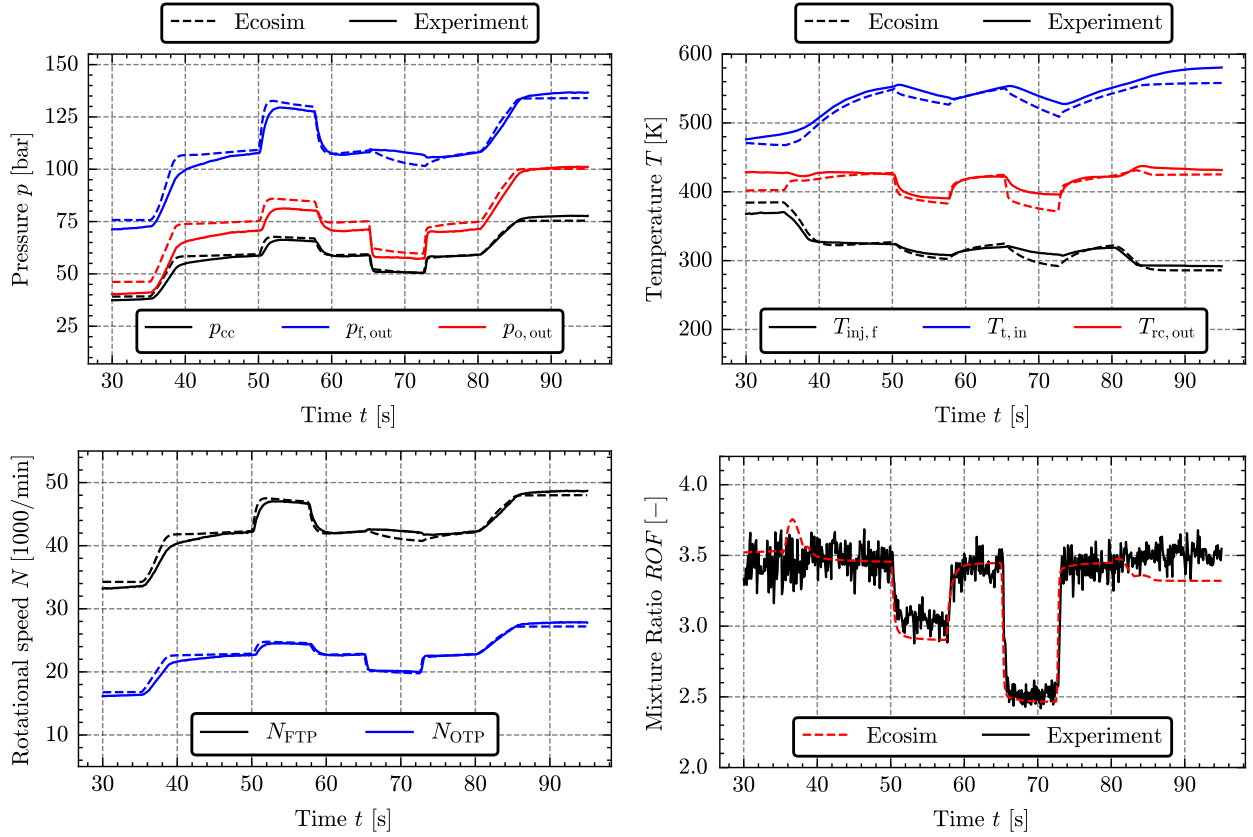


Figure 12: Comparison of the predictions of the LUMEN simulation model with experimental data. The compared values are the combustion chamber pressure p_{cc} , the fuel and oxidiser pump outlet pressure $p_{f,out}$ and $p_{o,out}$, the fuel injection temperature $T_{inj,f}$, the cooling channel outlet temperature $T_{rc,out}$ and turbine inlet temperature $T_{inj,f}$ and $T_{t,in}$, the rotational speed of the FTP and OTP N_{FTP} and N_{OTP} and the mixture ratio ROF .

well-suited for a wide range of applications. During the 2025 hot-fire test campaign, the model has been successfully used to generate a sequence replicating the thrust profile of the Apollo 15 Lunar Module during landing [8]. With this bi-directional validation, it can also be concluded that the model is well-suited for the development of intelligent controllers and health monitoring systems which are key of next generation rocket engines.

6. Summary and outlook

In this work the LUMEN simulation model which was initially developed by Dresia [6] using EcosimPro with the European Space Propulsion System Simulation (ESPSS) toolkit is updated and validated using experimental data from the latest test campaign conducted in 2025. For this, the components of the thrust chamber, cooling channels, injectors, turbines and pumps are built and validated individually using experimental data from the latest hot-fire test campaign of LUMEN. All components were modeled with high accuracy. After, all components are integrated into a complete LUMEN model. The complete model is compared against experimental data from a test run which covers the complete operational envelope of LUMEN with steps and ramps as transients. Over the entire test duration, the model demonstrated a high degree of accuracy with a mean average percentage error of 2.4 % and 3.5 % for the combustion chamber pressure and mixture ratio, respectively.

Despite its high accuracy, the simulation model is developed with remaining uncertainty in the torque characteristics for the FTP and OTP. In accordance with LUMEN's roadmap, test campaigns are scheduled with the purpose of acquiring data for the complete characterisation of the turbines and pumps of the FTP and OTP. Within future activities it is planned to compare the simulation model against experimental data of more test runs. In the subsequent phase of the project, it is planned to develop intelligent controllers and fault diagnosis systems using the simulation model. Those advanced technologies will be experimentally tested at the European research and technology test bench P8.3 in Lampoldshausen in the future, paving the way for next generation rocket engines.

References

- [1] W. Armbruster. *Untersuchung injektorgekoppelter Verbrennungsinstabilitäten in einer kryogenen Raketenbrennkammer*. Ph.d. thesis, RWTH Aachen, 2022.
- [2] E. Belyaev, V. Chevanov, and V. Chervakov. *Mathematical Modeling of the Work Process in Liquid-Propellant Rocket Engines (in Russian)*. MAI Moscow, 1999.
- [3] M. Börner, J. Martin, J. S. Hardi, D. Suslov, J. C. Deeken, and M. Oswald. Lumen Thrust Chamber - Flame Anchoring for Shear Coaxial Injectors. In *7th Edition of the Space Propulsion Conference*, Virtual, 2021.
- [4] R. H. Dos Santos Hahn, J. C. Deeken, M. Oswald, and S. Schlechtriem. Turbine Design and Optimization Tool for LUMEN Expander Cycle Demonstrator. *Journal of Physics: Conference Series*, 1909(1), doi: 10.1088/1742-6596/1909/1/012053, 2021.
- [5] R. H. Dos Santos Hahn, J. C. Deeken, T. Traudt, M. Oswald, S. Schlechtriem, and H. Negishi. LUMEN Turbopump - Preliminary Design of Supersonic Turbine. In *32nd International Symposium on Space Technology and Science*, Fukui, Japan, 2019.
- [6] K. Dresia. *Rocket Engine Control with Deep Reinforcement Learning*. Ph.d. thesis, RWTH Aachen, 2025.
- [7] K. Dresia, A. Adler, A. M. Saraf, R. H. Dos Santos Hahn, E. Kurudzija, T. Traudt, J. C. Deeken, and G. Waxenegger-Wilfing. Rocket Engine Control with Neural Networks: Experimental Results of the LUMEN Turbopump Test Campaign. In *AIAA SCITECH 2023 Forum*, National Harbor, United States of America, 2023.
- [8] K. Dresia, T. Traudt, M. Börner, D. Suslov, W. Armbruster, R. dos Santos Hahn, E. Kurudzija, C. Groll, M. A. Müller, S. Klein, J. Hämisch, J. Deeken, J. Hardi, and S. Schlechtriem. Hot-Fire Testing and System Analysis of the LUMEN Liquid Upper Stage Demonstrator Engine. In *3rd Int. Conf. on Flight Vehicles, Aerothermodynamics and Re-entry (FAR)*, Arcachon, France, 2025.
- [9] Empresarios Agrupados. EcosimPro 2024 Version 7.0 Complete Reference Manual, 2024.
- [10] Empresarios Agrupados and ESA. ESPSS 3.8.1 User Manual, 2024.
- [11] J. Haemisch, D. Suslov, G. Waxenegger-Wilfing, K. Dresia, and M. Oswald. LUMEN - Design of the Regenerative Cooling System for an Expander Bleed Cycle Engine Using Methane. In *7th Edition of the Space Propulsion Conference*, Virtual, 2021.
- [12] J. S. Hardi. *Experimental Investigation of High Frequency Combustion Instabilities in Cryogenic Oxygen-Hydrogen Rocket Engines*. Ph.d. thesis, The University of Adelaide, 2012.
- [13] J. S. Hardi, J. Martin, M. Son, W. Armbruster, J. C. Deeken, D. Suslov, and M. Oswald. Lumen Thrust Chamber - Injector performance and Stability. In *7th Edition of the Space Propulsion Conference*, Virtual, 2021.
- [14] E. Kurudzija, K. Dresia, J. Martin, T. Traudt, J. C. Deeken, and G. Waxenegger-Wilfing. Virtual Sensing for Fault Detection within the LUMEN Fuel Turbopump Test Campaign. In *9th Edition of the Space Propulsion Conference*, Glasgow, Scotland, 2024.
- [15] R. H. Dos Santos Hahn, K. Dresia. LUMEN Generic Turbopump Models. Technical report, DLR-LA-LM-TP-TB-005-01, Lampoldshausen, Germany, 2024.
- [16] D. Suslov, J. Haemisch, E. Zametaev, W. Armbruster, M. Börner, R. H. Dos Santos Hahn, S. Klein, K. Dresia, J. Hardi, T. Traudt, J. C. Deeken, and S. Schlechtriem. Development of a 25 kN Regeneratively Cooled LOX/LNG Thrust Chamber for LUMEN. In *9th Edition of the Space Propulsion Conference*, Glasgow, Scotland, 2024.
- [17] D. Suslov, J. Haubrich, J. Haemisch, K. J. Artzt, J. Munk, W. Armbruster, J. S. Hardi, and J. C. Deeken. Development and Qualification Testing of a Laser Powdered Bed Fusion Manufactured 25 kN LOX/LNG Regeneratively Cooled Thrust Chamber Produced from CuCrZr-Alloy. In *9th Edition of the Space Propulsion Conference*, Glasgow, Scotland, 2024.
- [18] T. Traudt, W. Armbruster, C. Groll, R. H. Dos Santos Hahn, K. Dresia, M. Börner, S. Klein, D. Suslov, E. Kurudzija, J. Haemisch, M. A. Müller, J. C. Deeken, J. Hardi, and S. Schlechtriem. LUMEN, the Test Bed for Rocket Engine Components: Results of the Acceptance Tests and Overview on the Engine Test Preparation. In *9th Edition of the Space Propulsion Conference*, Glasgow, Scotland, 2024.